Mohammed Adil Ali

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.93
NARAYANA JUNIOR COLLEGE	Class XII (Board of Intermediate Education, Andhra Pradesh (BIEAP))	2023	94.80%
NARAYANA (EM) SCHOOL	Class X (Board of Secondary Education, Andhra Pradesh (BSEAP))	2021	99.00%

PROFESSIONAL EXPERIENCE

Mobile Engineer Intern

Expedia Group | Bangalore, India (US Sort & Filter Team)

Tech Stack: Kotlin, Appium, WebDriverIO, UIAutomator, JavaScript, GraphQL, GitHub Actions, Browser Stack

- Engineered 100+ end-to-end UI tests for Android Sort & Filter components using Appium, UIAutomator, and WebDriverIO
- Built **robust** automated UI test suites covering **90%+ of Sort & Filter features** across multiple brand applications.
- Architected a **GitHub Actions** pre-merge gateway that automatically detects source code changes and executes **targeted UI test suites** to catch component failures prior to deployment, minimizing manual testing errors.
- Validated dynamic UI components against backend **GraphQL APIs** within Expedia's **Server-Driven UI Architecture**.
- Leveraged **responsible AI tools** to accelerate scripting while collaborating daily with the team and iOS co-intern.

SKILLS

Programming Languages: C, C++, JavaScript, Kotlin, Python

Tools & Technologies: Git, GitHub, GitHub Actions, Android Studio, Browser Stack, Appium, VS Code

Full Stack & API's: HTML, CSS, JavaScript, React, Node.js, Express.js, PostgreSQL, MongoDB, REST APIs, GraphQL

Soft Skills: Problem Solving, Team Collaboration, Critical Thinking, Adaptability, Time Management

RELEVANT COURSES

Computer Science Courses: Introduction to Programming, Data Structures & Applications, Operating Systems I, Operating Systems-II, Foundations of Machine Learning

Math Courses: Probability, Linear Algebra, Calculus, Differential Equations, Transform Techniques, Introduction to Statistics, Complex Variables

Management & Soft Skills Courses: Introduction to Entrepreneurship, Communication Skills, Ethics and Values

PROJECTS

DevWorkspace (Personal Project)

Tech Stack: React, Node.js, Express, Socket.io, Redis, PostgreSQL, JWT

- Engineered a distributed real-time chat platform delivering sub-millisecond messaging latency using React, Express, and Socket.io to enable seamless concurrent communication across channels.
- Scaled WebSockets horizontally using Redis Pub/Sub and cached message history via Redis Sorted Sets.
- Implemented asynchronous message persistence to batch database writes efficiently to Neon PostgreSQL.

URL Shortener & Analytics Engine (Personal Project)

Tech Stack: Node.js, Express, PostgreSQL, Redis, REST APIs, Base62

- Engineered a high-performance URL shortener delivering sub-15ms redirects using a Redis Cache-Aside strategy and Base62 encoding, significantly decreasing primary database read overhead.
- Built a custom sliding-window rate limiter using Redis Sorted Sets to protect core endpoints from request spikes.
- Designed an async analytics pipeline to queue click metadata and batch database writes to Neon PostgreSQL.

Multithreaded Synchronization Project (Course Project)

Tech Stack: C++, Pthreads, Semaphores, Locks

- Developed multithreaded Sudoku validators in C++ using static (Chunk/Mixed) and dynamic (TAS, CAS, Bounded CAS) task allocation methods to maximize multi-core processor efficiency.
- Implemented mutual exclusion algorithms and analyzed synchronization overhead across different approaches
- Solved the Producer-Consumer problem using semaphores and mutex locks while benchmarking the impact of synchronization on execution performance.

POSITIONS OF RESPONSIBILITY

Department Website Developer | Computational Engineering Dept.

- Coordinated with a team of developers to design and deploy the official responsive website for the department.
- Spearheaded front-end implementation using HTML, CSS, and JavaScript, managing UI design choices and component updates to deliver a seamlessly accessible portal for all department students and faculty members.

EXTRACURRICULAR & INTERESTS

- Won a state-level gymnastics medal and qualified for nationals through disciplined athletic training.
- Enjoy playing sports like cricket, football, table tennis, and basketball.
- Love following international esports tournaments and competitive gaming.